



GARDEN CLICKER



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Project Description:

This project aims to appeal to gamers who enjoy chill but fun idle games. Our game, "Garden Clicker" is a basic clicker game that takes the theme of flowers for a garden-like vibe. You progress through the game by getting upgrades which help gain more and more flowers. The game allows players to login, play for however long they desire, and they can leave the game on for however long they want, and the game will accumulate flowers passively.

Garden Clicker is a mobile game. It uses a login system that lets the players save their progress and load it whenever they want to play the game in the future. This means that players can open and close their game as they please and their progress will always be there. The main attraction of the game is the eagerness to grow their collection and attempt to beat the game by unlocking every achievement in the game.

Functional Requirements:

Authentication:

- Users can register with email and password (with confirm password validation)
- Users can log in with email and password
- Login requires passing a 2FA email verification code
- Users can request a password reset email
- Users can play as a guest (no account required, no cloud save)
- Users can log out from the settings screen

Core Gameplay:

- Tapping the flower increments the flower count by the current click power
- Passive income automatically adds flowers every second based on owned upgrades
- Flower count is displayed live in the HUD

Shop & Upgrades:

- Users can open/close the shop overlay from the game screen
- Users can purchase upgrades (Tulip, Flower Stand, Sakura's Blessing, Fountain of Growth) that increase passive income
- Users can purchase click power upgrades (Grass Touch, Water Touch, Sun Touch, Sun Power) that increase flowers per tap
- Each upgrade has a cost that increases per level purchased
- Rebirth discounts reduce shop upgrade costs

Rebirth System:

- Users can perform a rebirth when they have at least 1,000,000 flowers
- Rebirth converts flowers into rebirth energy (1 energy per 1M flowers)
- Rebirth resets flowers and regular upgrades but keeps rebirth upgrades
- Rebirth energy is spent in a separate rebirth shop on permanent upgrades:

- Dark Sun Touch — multiplies click power
- Flash Sale — discounts regular shop prices (15% or 25%)
- Time Warp — multiplies passive income
- Flower God — multiplies everything by 1000x

Achievements:

- The game tracks and unlocks achievements based on upgrade levels, rebirth milestones, and lifetime rebirth energy
- Unlocked achievements trigger an in-game toast notification
- Users can view all achievements and their unlock status on the Achievements screen
- A meta-achievement ("You Did It!") unlocks when all other achievements are earned

Save System:

- Signed-in users' game state (flowers, upgrades, rebirth, achievements) is saved to Firestore
- The game auto-saves every 30 seconds
- Users can manually save via the save button in the app bar
- The game loads the user's save on login and on app start
- Deleting a save wipes cloud data and resets the local state
- Guests never trigger cloud saves

Navigation:

- Main menu provides entry to the game or login
- From the game screen users can navigate to: Shop, Rebirth, Achievements, Settings
- Settings screen allows logout and save deletion

User Management:

- User's accounts are saved in an authenticator in the Firebase database.
- The user's progress is also saved in a Firestore that connects to the authenticator in the Firebase to grab the user's data from their email.
- The game automatically saves the gamestate every 30 seconds and stores their progress in the database.

Non-Functional Requirements:

1. Performance

- The game loop runs via Flame's update(double dt) — passive income is calculated every second using an accumulator, so it never blocks the UI thread.
 - ValueNotifier is used throughout (flowers, rebirth energy, change ticks) so only widgets that depend on a value rebuild — not the entire tree.
- Firestore writes are batched into a single set() call per save, not multiple document writes.

2. Reliability

- Auto-save runs every 30 seconds (both via the Flame game loop and the Timer.periodic in GameScreen).
- Save errors are caught silently with debugPrint — the game won't crash if Firestore is unavailable, but the user also won't be notified of a failed save.
- _suppressSave flag prevents a deleted save from being accidentally restored by a pending auto-save.

3. Security

- Authentication is handled entirely by Firebase Auth — no passwords are stored locally.
- Two-factor authentication (2FA) is enforced via email code on every login.
- Passwords must be confirmed on registration.
- Forgot password sends a Firebase reset email.
- Save data is keyed by user email in Firestore (saves/{email}), meaning users can only access their own save.

4. Usability

- The login screen toggles between login/register in one place — no separate screens.
- Password visibility toggles are provided on all password fields.
- Error and success messages are color-coded (red for errors, green for success).
- The shop, rebirth, and achievements are accessible from the main game screen without leaving it (overlays/navigation).
- Achievement toasts appear in-game without interrupting gameplay.

5. Compatibility

- Built with Flutter, so it targets Android, iOS, macOS, and Chrome from a single codebase.

- Currently confirmed working on Android (emulator-5554, API 36) and Chrome.
- iOS/macOS is not yet fully functional due to missing Xcode setup.

6. Maintainability

- Upgrades, rebirth upgrades, and achievements are defined in separate `_definitions.dart` files — adding new content only requires editing those files, not the managers.
- Managers (UpgradeManager, RebirthManager, AchievementManager) are cleanly separated with their own serialization (`toJson/fromJson`).
- Achievement checks are idempotent — safe to call multiple times without side effects.

7. Scalability

- Firestore scales automatically — no server to manage.
- Each user has one document in the `saves` collection, so the data model is flat and efficient.
- Adding new upgrades or achievements requires no schema changes — the JSON is flexible.

8. Accessibility

- No explicit accessibility features are implemented (no semantic labels, no screen reader support, no font scaling).
- This is a current gap — Flutter supports Semantics widgets but they haven't been added.

9. Data Integrity

- Save data uses `SetOptions(merge: true)` so a partial save won't wipe fields that weren't included.
- `lastSaved` is written as a server timestamp (not the device clock), so it's accurate across timezones.
- On auth state change, the game fully resets and reloads from the new user's save — no data bleed between accounts.

Roles and responsibilities

Kiara's contributions:

1. Front-End development

- Designed and implemented user interface using Flutter widgets
- Built layouts that are responsive for different screens
- Made sure UI was consistent across all screens
- Game styling like: colors, theme, fonts, icons, buttons, spacing

2. Back-End - These are features that I just worked on the code for

- Splash screen
- Settings
- Play button
- Quit button

Also for these features of back and front-end, I implemented the UX design to make sure the buttons changed screen smoothly, improved usability and accessibility, and made sure the actions were easy to understand and get good feedback on.

For documentation, I did help out with the first powerpoint we did, I did the second powerpoint, and I am helping out with this documentation.

Firestore

I started the firestore code just like we did in class. I followed what google said to do and implemented the code where it said to be implemented.

We used this:

<https://firebase.google.com/docs/auth/flutter/start>

<https://www.youtube.com/watch?reload=9&v=rWamixHIKmQ>

Authentication

- Login
- Registration
- Forgot password
- Two-factor authentication

Details:

- Creating login forms
- Input validation
- Error handling
- Password reset workflow
- Connecting frontend forms to Firebase authentication

For the core authentication system, Firebase Authentication was used to take care of our users accounts. This allowed users to create an account, log in with their email and password, and

have authenticated sessions. Firebase took care of storing users information, authentication validation, and session management.

Forgot password

If a user forgets their password they don't have to create a new account. Using Firebase's password recovery system, users could request a reset password directly from the login screen, receive a password reset link in their email, and create a new password to get access to their account.

2FA

Using EmailJS, I did research and it was quite easy to get the public API key, serviceID, and TemplateID. After a user successfully logged in through Firebase, the application asks for a second verification. The system generated a one-time verification code, which was sent to the user's email using EmailJS(lets the app send emails from the frontend without our own backend server). The user then enters their code on the two-factor authentication screen, and only after successful verification were they allowed into the main application.

This security flow:

- User enters email and password
- Firebase validates login credentials
- If authentication succeeds, a verification code is generated
- EmailJS sends the code to the user's registered email
- User enters the verification code on the 2FA screen
- Access is granted only if the code is correct

7. Planning(ideas, thoughts, plans)

- Game features
- Functionality decisions
- App flow
- Design
- App decisions like if we include: 2FA, rebirth, achievements, settings

Oliver's contributions:

1. Front End Development:

- Designed the main game, splash screen, upgrades view, rebirth menu, achievement menu, leaderboard menu.
- All sprites were handmade by me using Aseprite on Steam.

2. Back End Development:

- I managed the backend for the splashscreen after kiara gave me the blueprint
- I coded the backend for upgrades, rebirths, main flower, game save, load state, leaderboard, achievements, and connectivity for firebase connection to user data.

More

For the front end, I implemented things like having upgrades be in a gridview so that the user can toggle the upgrades so that they can decide if they want to see the full main game or focus on getting upgrades. Also, having the Rebirth count was for the convenience for the user so that they are able to see the specific amount of rebirths they can get even if they are in the billions (Billions = not able to see exactly how many millions you have). Also having the achievements displayed when earned is another factor for UX because many people love the feeling of obtaining achievements, and seeing it popup is very satisfying for many gamers.

For the backend UX, having the game automatically save every 30 seconds was for the user to have the ability to save their progress without having to constantly press the save button. Also, it prevents data loss on crash or power outages since the data is stored in our database. Also having a leaderboard give the user a competitive kick to try and reach the number 1 spot in our game.

Documentation

For documentation, I did most of the first powerpoint and I wrote about half of the final documentation.

Firebase

I used Firebase for users to save their game, and the leaderboard requires data from the database to work.

Database connectivity

How the application connects to a database to store, retrieve, and manage user data. We used Firebase, the cloud backend database. What the connection does:

- store user account information
- Gets saved user data
- login sessions
- save player progress
- sync achievements and upgrades
- manage personalized users data
- Loads leaderboard by ordering users by flowers.

When the app starts, Firebase is initialized: `await Firebase.initializeApp();`

This makes the connection between the Flutter application and the Firebase project.

The app then communicates with Firebase using Firebase services like:

- Firebase Authentication: for user account data
- Cloud Firestore/Firebase Database: for saving and getting back game progress

Example:

- User logs in
- Firebase validates data
- Firebase returns the authenticated user
- App gets that user's saved game data
- Progress is showed in the game
- Any changes are updated back to the database

User feedback

We took user feedback very seriously since it is a game and we need users to be successful. We used the feedback to improve the application's usability, functionality, and overall experience. After many tests, the feedback gave us direction to identify areas that needed improvement, such as navigation flow, interface clarity, and feature accessibility. We might not see some flaws, so with constructive criticism we made changes.

Examples of feedback:

- Improving screen navigation to make movement between login, main menu, gameplay, settings, and achievements more intuitive
- Layout: to create a cleaner and more visually organized design
- Authentication: adding features like forgot password recovery and clearer login validation messages
- Adding two-factor authentication to improve account security
- Improving feedback messages and alerts so users better understood actions, errors, and system responses

Seeking guidance when faced with difficulties

During the process we needed to search for help a lot.

Examples:

- Teammates and each other to troubleshoot, divide complex tasks, and review
- Researching documentation and online resources for technologies such as Flutter, Firebase, EmailJS, ChatGPT, ClaudeAI
- Testing different solutions through debugging
- Past projects

When using EmailJS, and Firebase, it needed understanding third-party services, API configuration, and secure workflow design. We had to do research on it since it was the first time using it and researched on google and ChatGPT how to implement and if it was a good suit for what we needed. (<https://dashboard.emailjs.com/admin/account>)

Structures and Algorithms:

We decided on our structures and algorithms based on what we needed for the game to keep it as clean as possible.

For data management, we used Firebase's document-based cloud storage for dealing with authentication, saved progress, achievements, upgrades, and settings, since this information needed to be accessed, updated, and synchronized.

In Flutter, collections like lists, maps, and objects/classes were used to organize data and UI.

For example:

- Lists were used to keep achievements, upgrades, or game items
- Maps and key value structures were used for storing paired data such as user settings, item values, or upgrade costs
- Custom classes/objects were used to model structured entities and improve code organization

For algorithms, we chose simple functionality like:

- Authentication to verify login credentials
- Two-factor authentication verification algorithm for verification codes
- Game progression algorithms for updating scores, currency, upgrades, and rebirth calculations
- Dynamically updating the UI when data changed
- Data retrieval and persistence operations for saving and loading user progress

For game-like features, we used FlameGame structure in our code for things like the game loop that provides the Update(dt) method that drives the passive income and auto save feature. Many classes use Flame like the class for the main clickable flower, passive income system and overlay system.

User Stories

Title: User Registration

As a user

I want to be able to create my account using my username, email and password

So that I can save my progress

Title: Login

As a user

I want to login using my email and password

So that I can access my saved account

Title: Forgot Password

As a user

I want to reset my password if I forget it

So that I can regain access to my account without creating a new one

Title: 2-Factor Authentication

As a user

I want to verify who I am using the code sent to my email

So that my account can have an extra layer of security

Title: Gameplay

As a user

I want to play the clicker game

So that I can earn flowers to advance

Title: Upgrades

As a user

I want to be able to buy upgrades using my flowers

So that I can progress the game faster and unlock new powers

Title: Achievements

As a user

I want to be able to view all the achievements

So that I can see my progress and accomplishments

Title: Rebirth

As a user

I want to be able to rebirth

So that I can unlock the permanent rebirth upgrades

Title: Save Progress

As a user

I want my games progress to be saved

So that I do not lose my achievements, upgrades, or flowers, and so I can show up on the leaderboard

Title: Load Progress

As a user

I want to load my previous game save data

So that I can continue where I left off

Title: Settings

As a user

I want to view settings

So that I can return to main menu, logout/in, reset progress and exit the game

Title: Logout

As a user

I want to be able to logout of my account

So that I can ensure no one can mess with my data from my computer.

Title: Leaderboard

As a user

I want to view the leaderboard

So that I can see who has the most flowers

Title: Profile

As a user

I want to view my profile

So that I can change my username.

Test cases

1. Registering a new user

Test: Create a new account

Steps:

1. Open the app
2. Go to login then register
3. Enter email and password
4. Press Register

Result:

User account is created and user goes to login/verification flow.

2. Login

Test: Log in with valid credentials

Steps:

1. Enter registered email
2. Enter correct password
3. Press Login

Result:

Firebase allows the data and the user continues to 2FA verification.

3. Not right credentials to login

Test: Log in with wrong password

Steps:

1. Enter valid email
2. Enter incorrect password
3. Press Login

Result:

An error message and the user is not logged in.

4. Forgot password

Test: Reset forgotten password

Steps:

1. Click Forgot Password
2. Enter registered email
3. Submit request

Result:

Firebase sends a password reset email link to the user.

5. 2FA

Test: Make sure correct login using email code

Steps:

1. Log in successfully
2. Check email for verification code
3. Enter the right code in the 2FA screen
4. Press Verify

Result:

User gets into the main game.

6. Not right 2FA code

Test: Puts in wrong code

Steps:

1. Log in successfully
2. Enter an incorrect 2FA code
3. Press Verify

Result:

Game shows an error and no access is granted.

7. API call to EmailJS

Test: Send 2FA email

Steps:

1. Complete Firebase login
2. Trigger 2FA code sending

Result:

EmailJS sends the verification email.

8. Click function

Test: Get currency by clicking alot

Steps:

1. Open the game screen
2. Press the flower

Result:

Flower count increases.

9. Buying upgrades

Test: Buy an upgrade

Steps:

1. Earn enough flowers
2. Open upgrade/shop menu
3. Choose an upgrade

Result:

Flowers decrease and upgrade is granted.

10. Not enough flowers

Test: Tries buying upgrade without enough flowers

Steps:

1. Open upgrade/shop menu
2. Select an upgrade that costs more than what you have

Result:

Purchase is denied

11. Saving progress

Test: Save game data

Steps:

1. Earn flowers
2. Buy an upgrade
3. Close or leave the app

Result:

Progress is saved to Firebase/cloud storage.

12. Loading previous progress

Test: Load saved game data

Steps:

1. Reopen the app
2. Log in with same account used before with saved data

Result:

Saved flowers, upgrades, achievements, and rebirth.

13. Achievements

Test: Unlock achievement

Steps:

1. Do what is needed to obtain an achievement
2. Check for achievements

Result:

Achievement shows unlocked.

14. Rebirth

Test: Use rebirth system

Steps:

1. Reach rebirth requirement
2. Click rebirth option
3. Confirm rebirth

Result:

Progress restarts and bonus/rebirth level increases.

15. Logout

Test: Log out safely

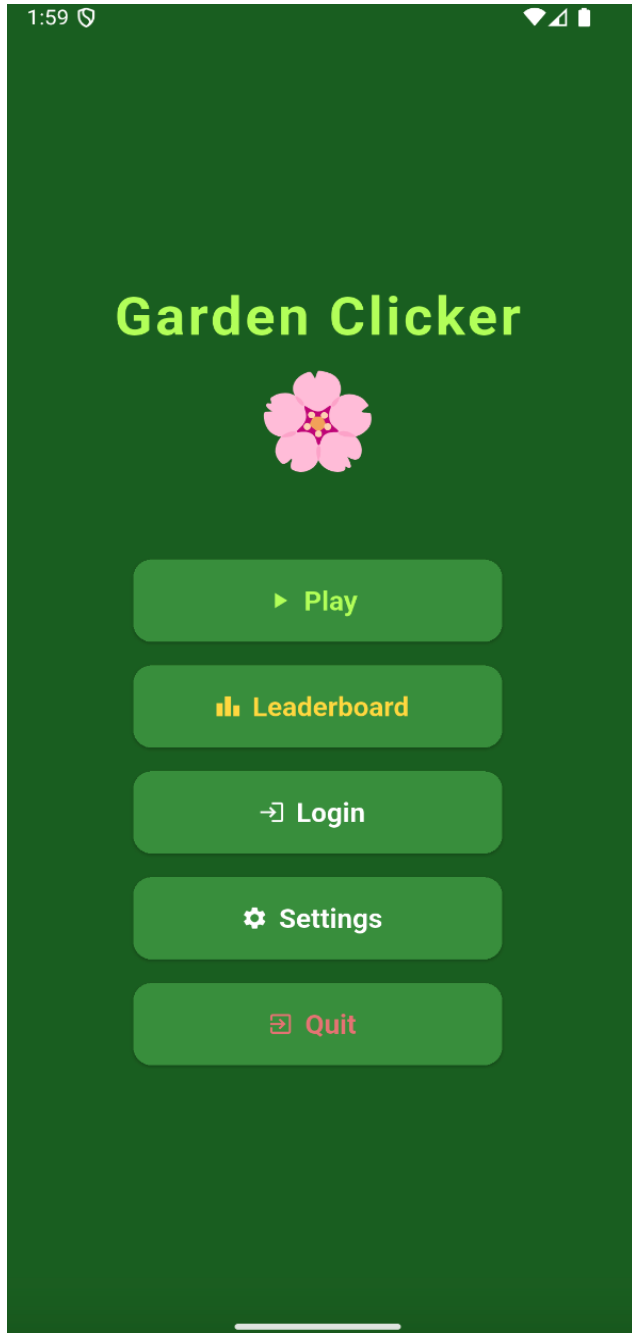
Steps:

1. Open settings/menu
2. Click Logout

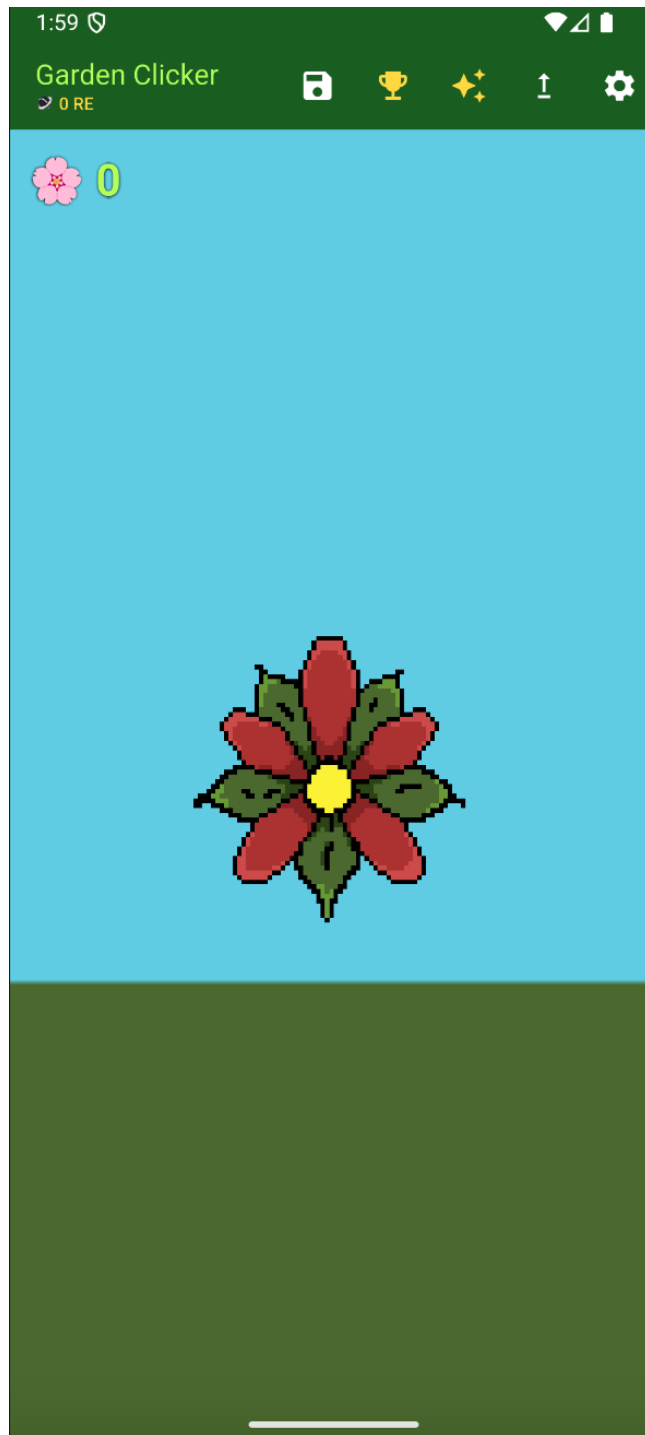
Result:

User is signed out and sent back to login.

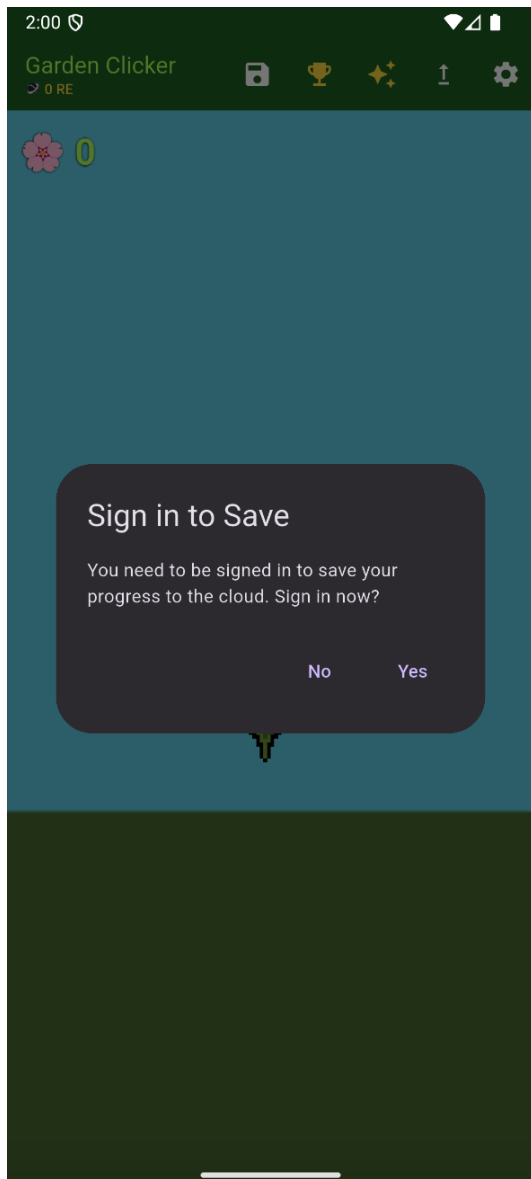
User Guide:



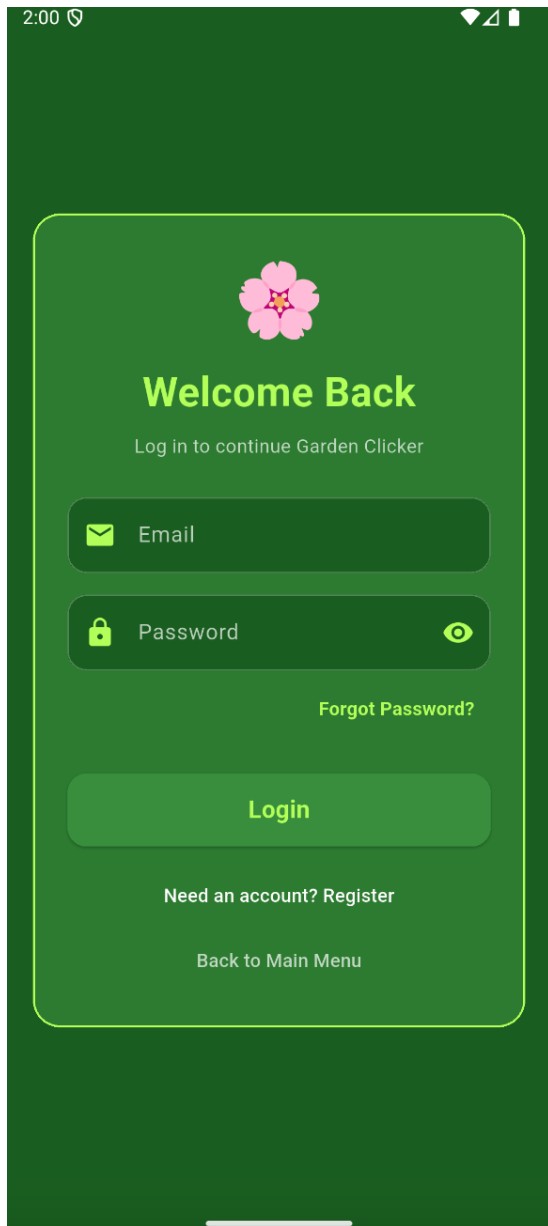
Welcome to Garden clicker! This will be your personal guide to understanding how Garden Clicker works and how you can reach the top! For starters, this will be the screen that renders when you first ever open the game.



If you decide to press play right away, this would be the beginning of your journey in collecting all the flowers. However, there is an issue here, you have not logged in yet. As a guest user, you can play the game as you wish, however, you will be missing out on important features. Most importantly, you will be unable to save your progress, so you will have to complete the game in one sitting, and that is not recommended.



This is an example of losing out on the save feature, so let's go ahead and fix that. When you press the save button on the right of "Garden Clicker", this will show up on your screen. Pressing "yes" will redirect you to the login page which is recommended. Let's Go!



This is the login page, the main page for users that already have an account with us, however, since you're new, we are going to have to register an account first. To do so, press "Register" and it will redirect you to the register page.

2:00



Create Account

Register first, then log in

Email

Password

Confirm Password

Username

Register

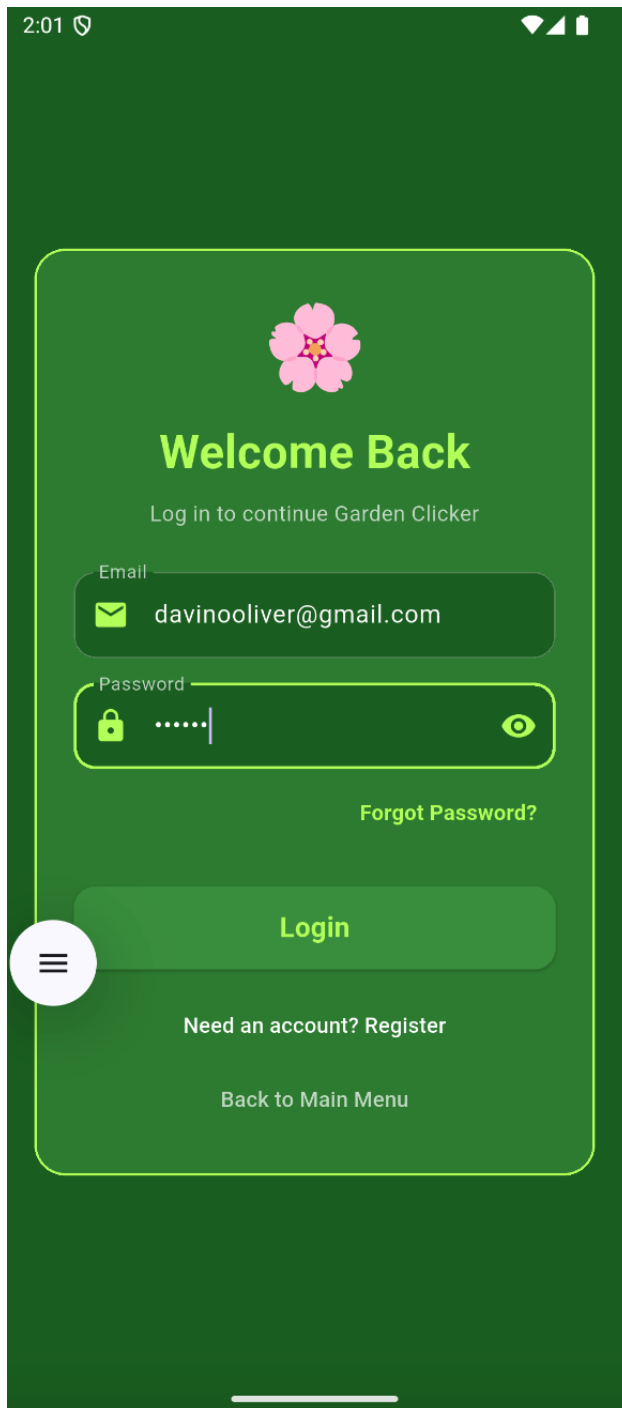
Already have an account? Login

Back to Main Menu

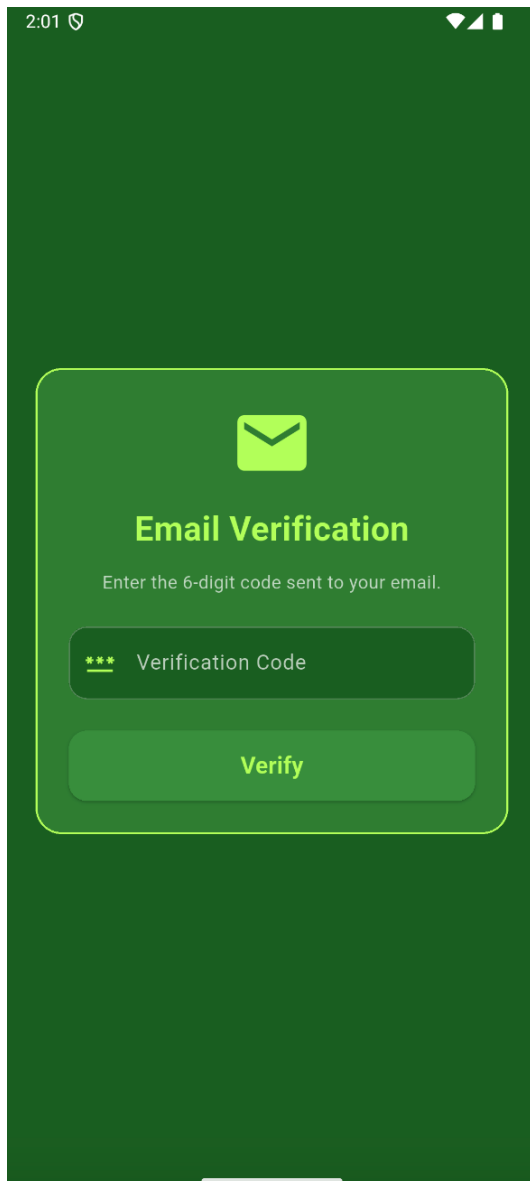
To create an account, you'll need to enter 4 boxes:
email, password, confirm password, and username.

So go ahead and enter your email, password, and
username so that your account can get registered.

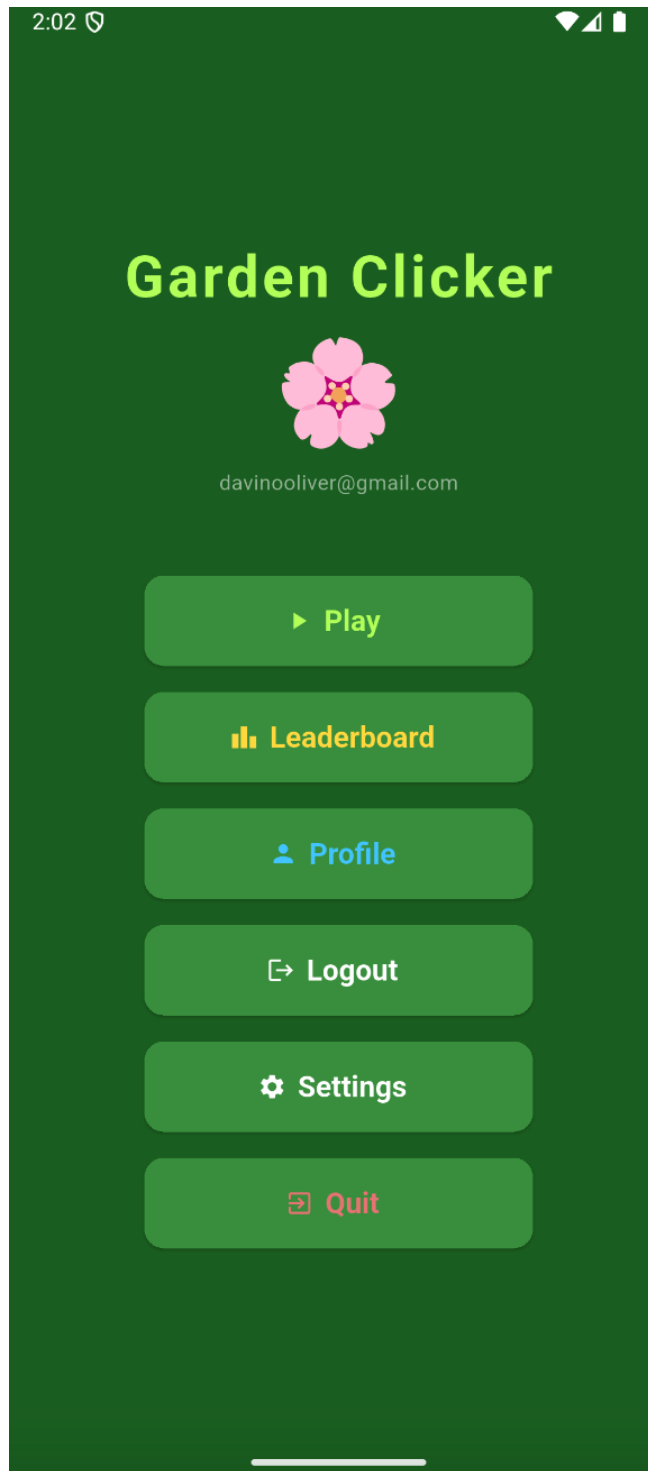
FYI, username and email can only be used once, so
don't steal another player's username or email
because it won't work! Anyways, let's move on.



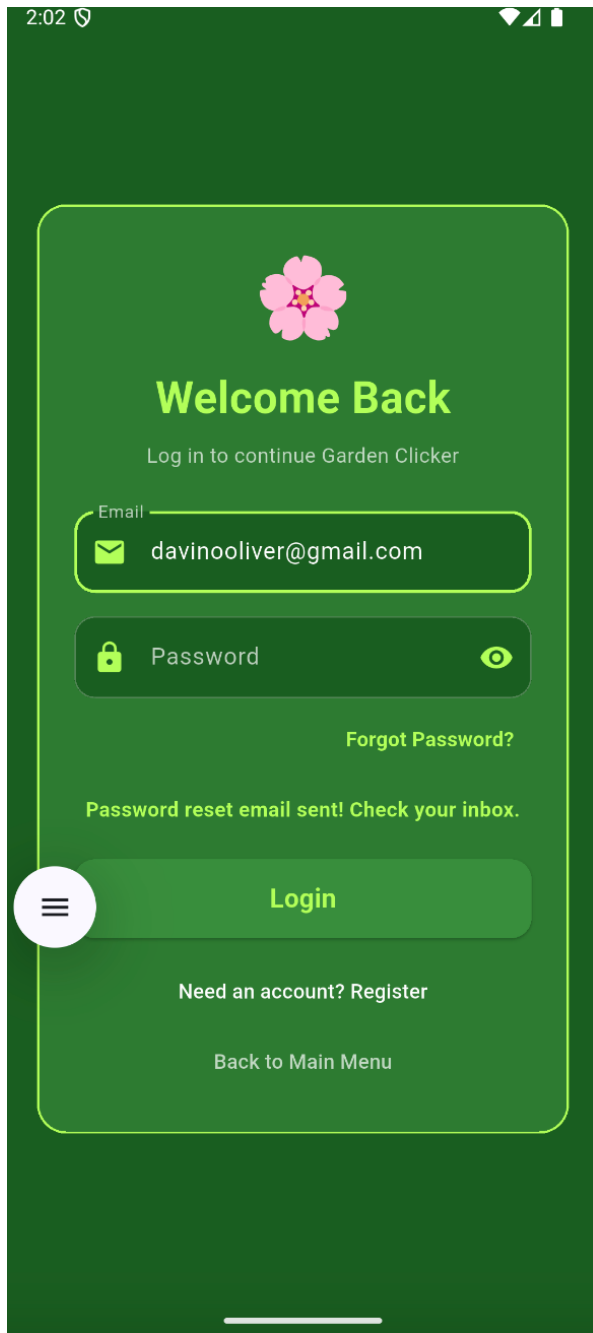
Once you've created your account, go ahead and enter your username and password in the login screen. Make sure the inputs are the exact same as the ones you entered in the register page. Once you are confident in your password, go ahead and Login!



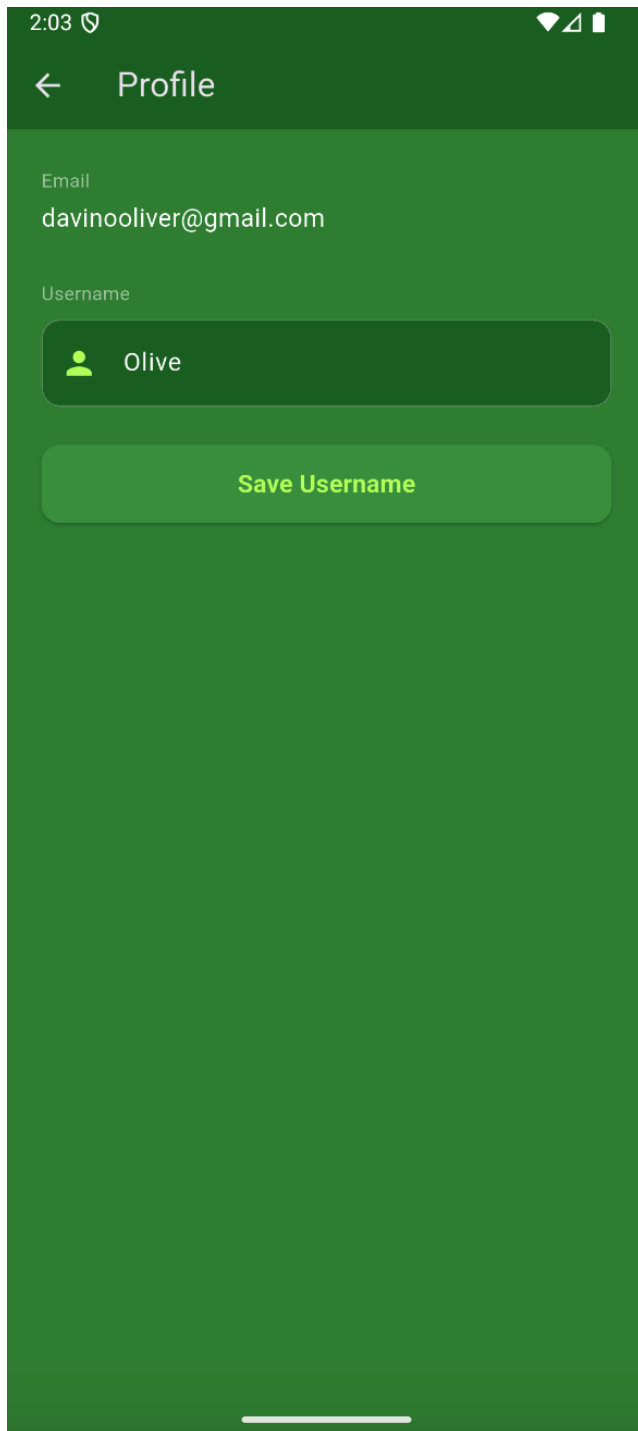
Don't forget about safety! When you attempt to login, for safety purposes, we send your email a verification code so that if someone else tries to login to your account, you will 1. Get notified, and 2. Be protected by 2-factor authentication. Once you get your code, enter it in the input and continue.



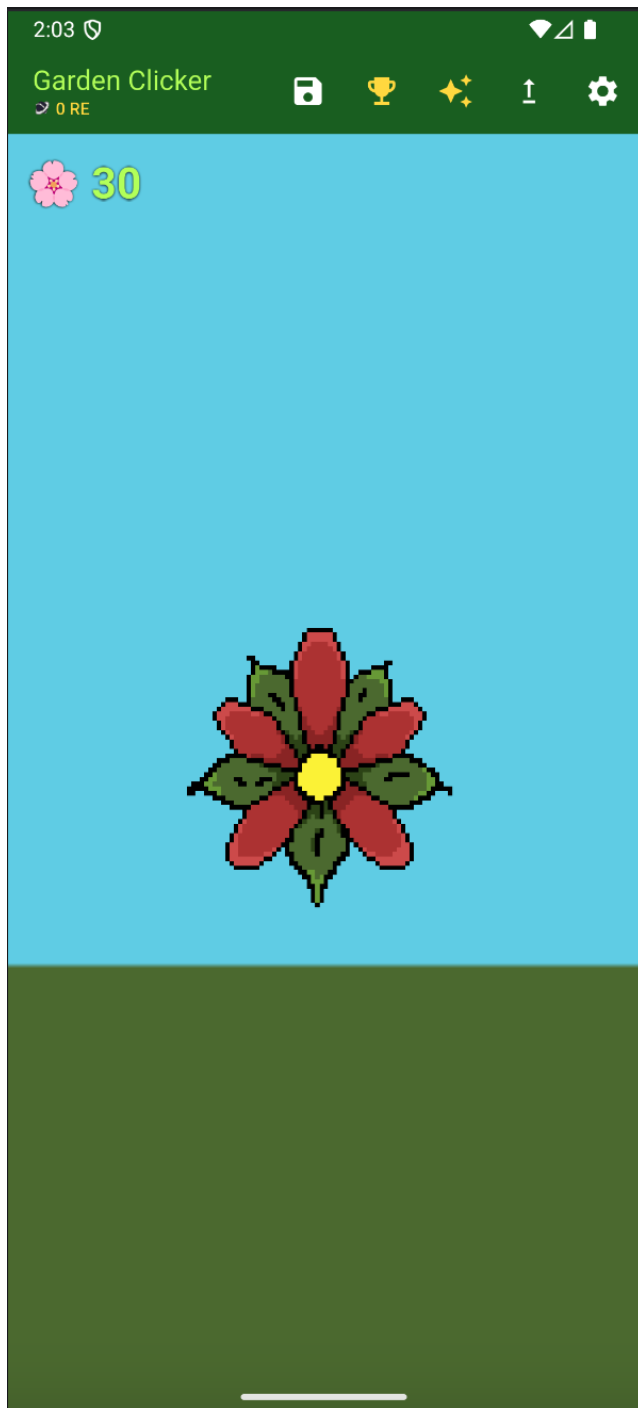
You did it! Your account is officially made, and you have logged in to the game. Now, you can view your profile and logout whenever you desire.



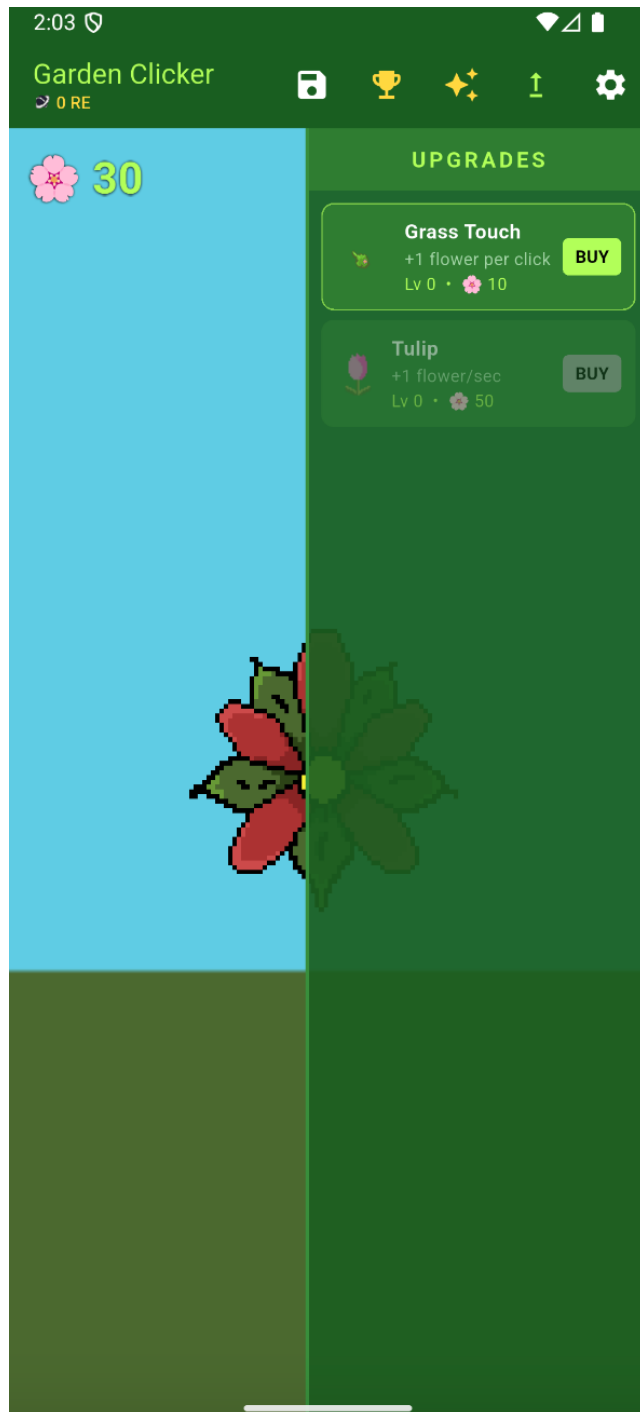
But Wait!!! Just in case you forget your password, we give you the option to change it. Just press “Forgot Password?” and you will get an email which prompts you to change your password, once that is done, you can enter the new password and continue.



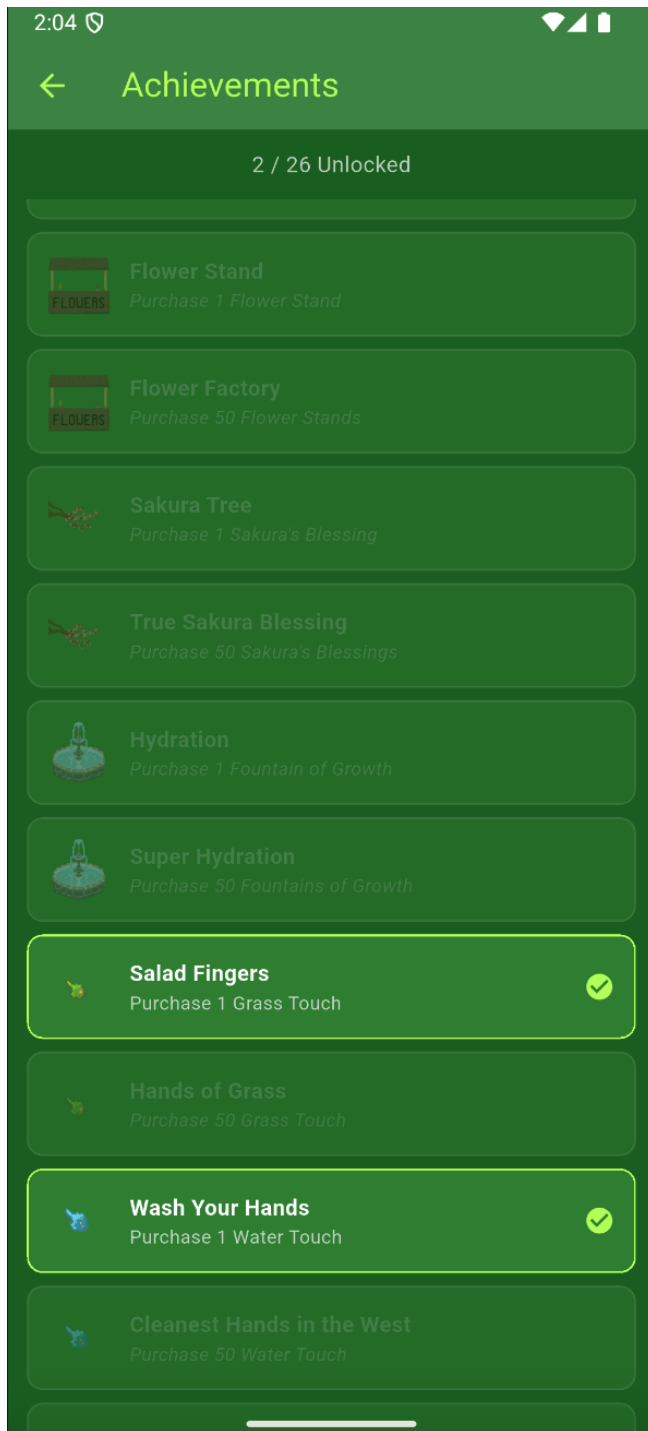
Anyways, back to profile. Here, you see your email and username. If you want to change your username, go ahead and enter the new username you want and press “Save Username”. As long as someone else does not own that username, it shall save as your new username. For this account, let’s go with “Olive”.



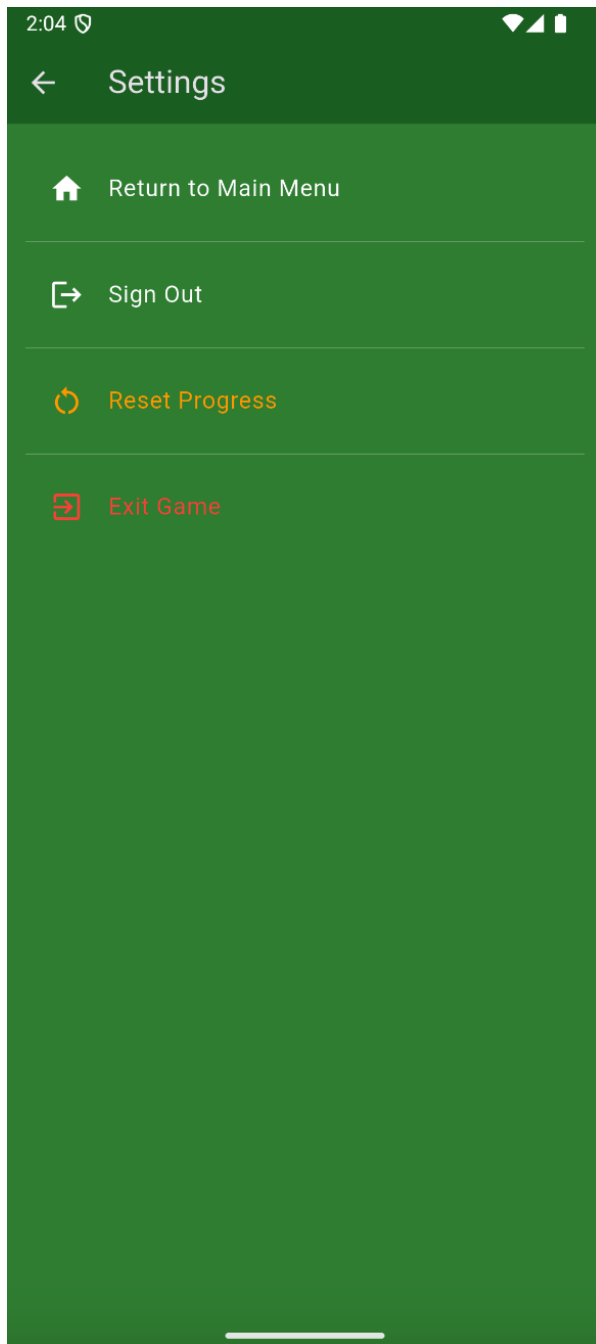
Let's finally get on with the game. Here is what it looks like once you press play and are logged in. Each time you click the flower, you increment the counter on the top right by 1. This is the beginning of your journey to unlocking everything!



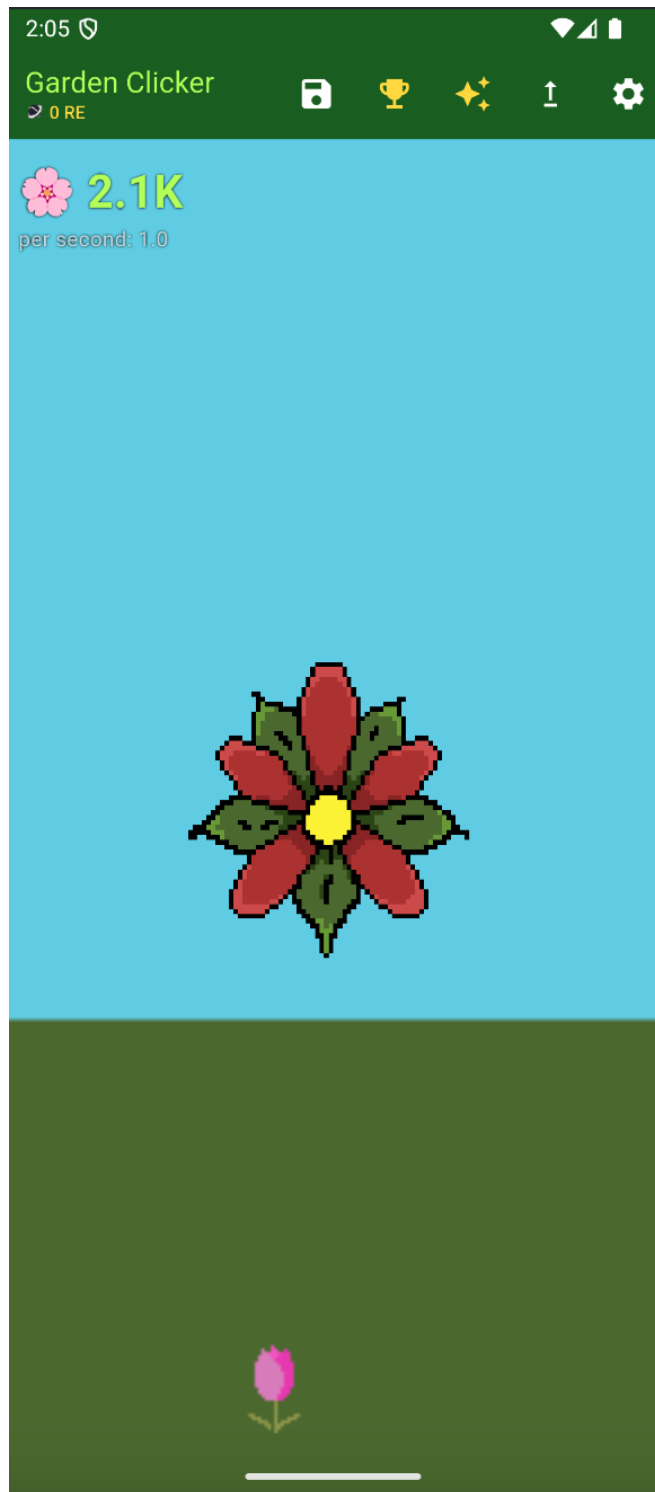
To toggle the upgrades tab, click the arrow icon next to the gear icon on the top right. This will display all the upgrades available. Yes, I know it looks empty, but that's simply because you haven't gotten enough flowers yet. The more upgrades you get, the faster your garden will grow, and you will gain more flowers per second. Let's move on to achievements!



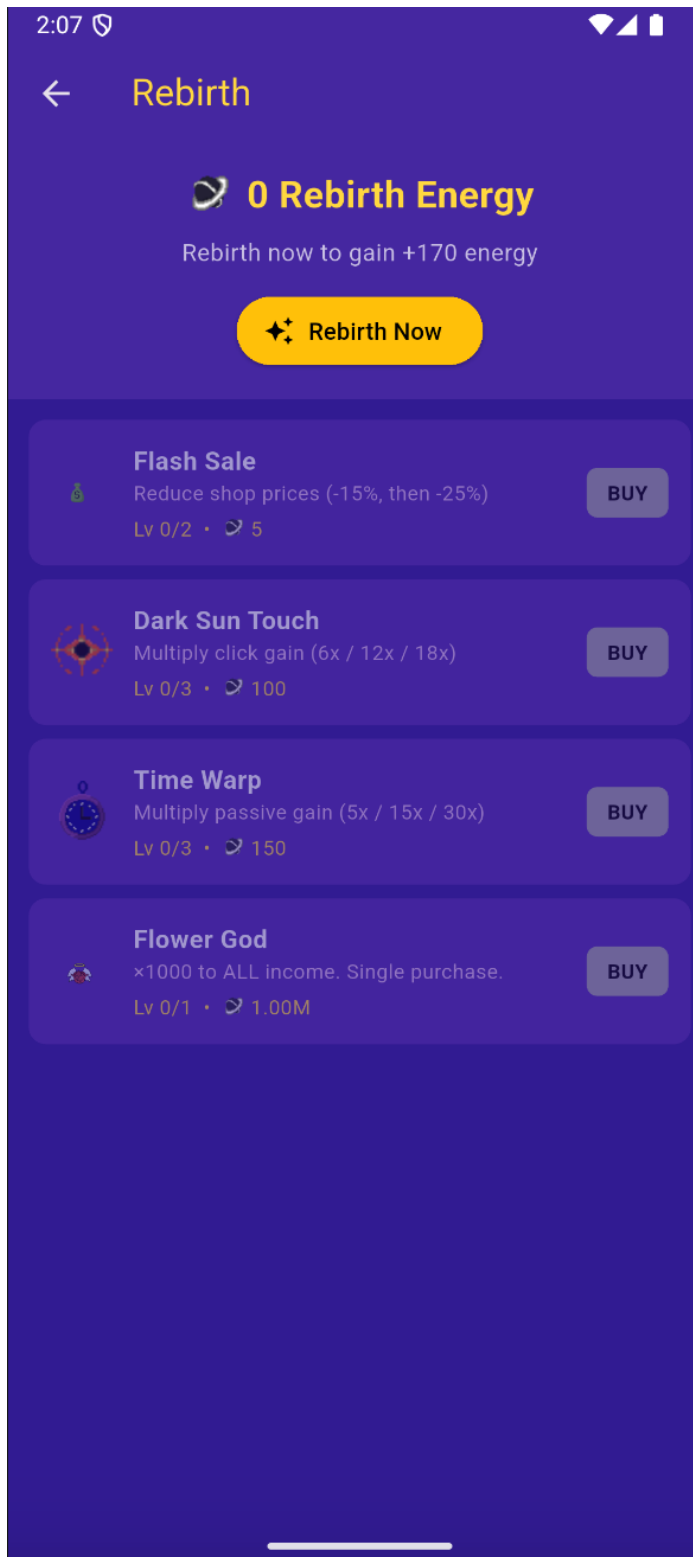
When you press the achievement tab (the trophy), you will open this menu that display all the achievements in the game and how to unlock them. The achievements are based on upgrades and other things. And whenever you meet the requirements of getting an achievement, it will display on the top of your screen.



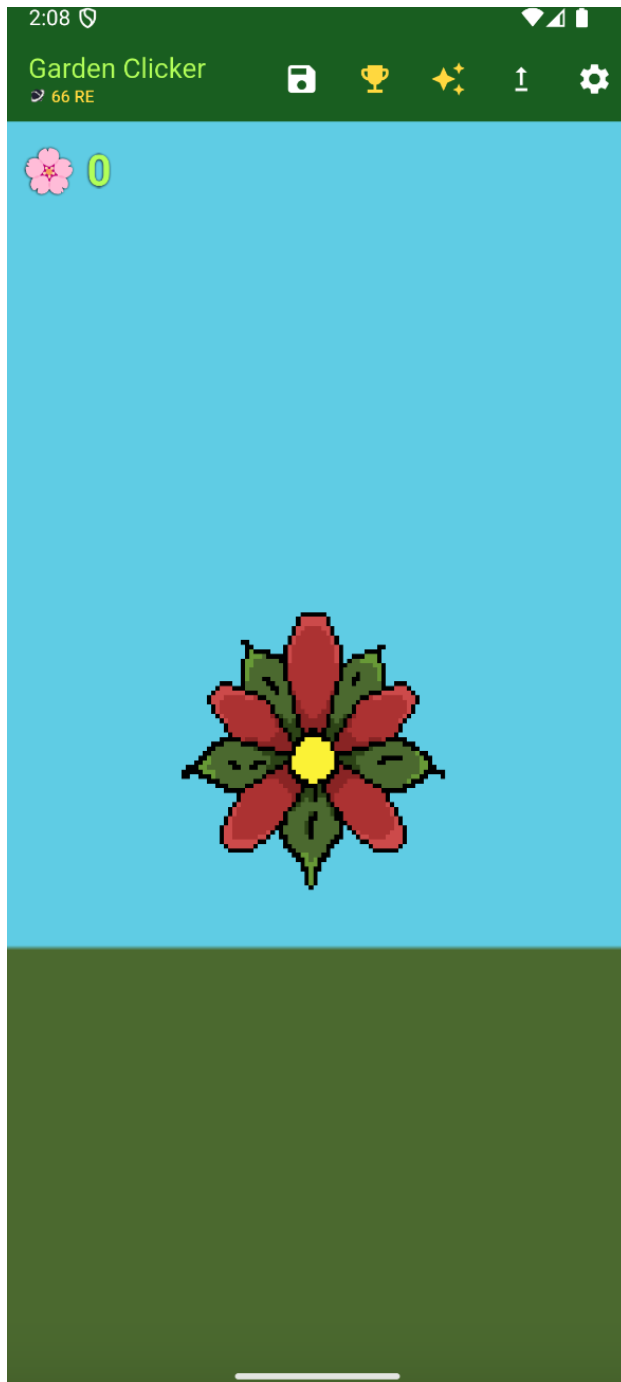
This is the settings menu, it allows you to return to the main menu, sign out, reset progress, and exit game. It goes without saying but be careful with the reset progress button. It does exactly what it says, returns to absolute 0.



When you progress and purchase certain upgrades, they will actually appear in your garden, and a “per second” will display under your flower count, letting you know how many flowers your passive income makers are generating. This is what your first tulip looks like!



Now here is for the more late game players. If you have already unlocked all the upgrades, or have millions of flowers to spare, you can rebirth for a new variety of upgrades. When you rebirth, you gain rebirth energy which is the new currency for rebirth upgrades. 1 rebirth energy is worth 1 million flowers.



Once you rebirth, your flower count and all your upgrades will be completely reset, with the exception of the permanent upgrades that were purchased in the rebirth menu. Now, there are some interesting features that come with rebirthing but to avoid spoilers, you should go ahead and give the game a try. Let us know what you think!



Now you know how to play our game and reach the top! To truly reach the top, the leaderboard is here to show you who truly reigns above everyone else. Go ahead and do your best and try to reach the very pinnacle of flowers! Good luck and have fun!